

Readme Enterprise AI Knowledge Assistant

Project Type: LLMOps / Retrieval-Augmented Generation (RAG) Proof of Concept

Deployment: Dockerized FastAPI + local Ollama + local Streamlit

Observability: Prometheus + Grafana + Opik

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Project Overview

The Enterprise AI Knowledge Assistant is an LLM-powered Retrieval-Augmented Generation (RAG) application that allows users to ask natural-language questions about a collection of enterprise documents.

Instead of asking a Large Language Model (LLM) to answer from its general training knowledge, the application first retrieves relevant document chunks from an enterprise knowledge base and then supplies those chunks to the LLM as context.

The application is designed as an LLMOps-oriented proof of concept and demonstrates:

- Enterprise document ingestion
- Document chunking
- Semantic vector search
- FAISS-based vector storage
- Hugging Face sentence embeddings
- Local LLM inference through Ollama
- FastAPI REST API
- Streamlit user interface
- Docker-based deployment
- Prometheus application metrics
- Grafana operational dashboards
- Opik LLM/RAG tracing
- Application logging
- Automatic FAISS index rebuild when source documents, embedding model, or chunking configuration changes

Key Features

RAG

- PDF document ingestion
- Recursive character-based chunking
- BGE-small embeddings
- FAISS vector index
- Top-4 semantic retrieval
- Context construction
- Grounded LLM generation

API

FastAPI endpoints include:

- /
- /health
- /version
- /chat
- /reindex
- /test-llm
- /metrics

Interactive API documentation is automatically provided by FastAPI at /docs.

User Interface

Streamlit provides:

- Question input
- Sample questions
- Ask button
- Clear button
- Generated answer
- Source documents
- Expandable metadata

- Raw JSON response
- Links to Grafana, Prometheus, and Opik

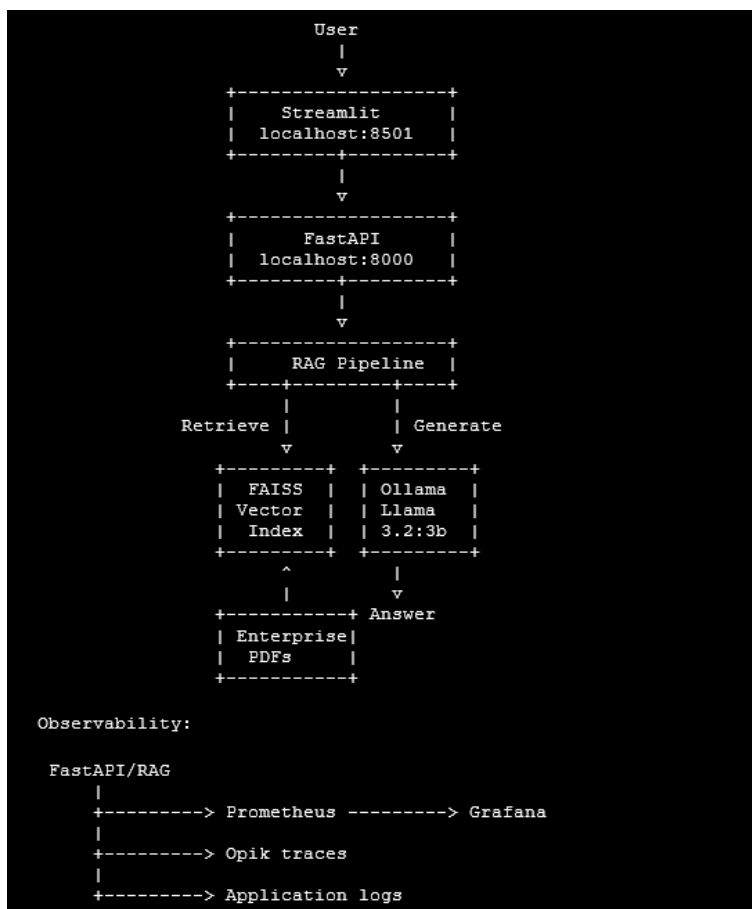
Streamlit is intentionally kept outside Docker for development convenience.

Observability

The application uses three complementary observability layers:

Tool	Purpose
Prometheus	Collect application metrics
Grafana	Visualize operational metrics
Opik	Trace individual RAG/LLM requests

High-Level Architecture



Technology Stack

Layer	Technology
Programming language	Python 3.11
API	FastAPI
RAG orchestration/utilities	LangChain
Vector store	FAISS
Embeddings	BAAI/bge-small-en-v1.5
LLM	llama3.2:3b
LLM runtime	Ollama
UI	Streamlit
Containerization	Docker / Docker Compose
Metrics	Prometheus Client
Metrics collection	Prometheus

Visualization	Grafana
LLM tracing	Opik
Logging	Loguru
Configuration	Pydantic Settings
Dependency management	uv

Observability

Observability is connected to the application independently:



Knowledge Base

The current FAISS index metadata records six PDF source documents:

1. BI_Glossary.pdf
2. Business_FAQ.pdf
3. Business_Policies.pdf
4. Company_Overview.pdf
5. Northwind_Annual_Report_2025.pdf
6. Northwind_IT_Policy.pdf

The stored index metadata records:

Setting	Value
Source documents	6
Indexed chunks	55
Chunk size	1000
Chunk overlap	200
Embedding model	BAAI/bge-small-en-v1.5

The FAISS index is stored under: **data/faiss/index/** with **index.faiss** and **index.pkl**

and index metadata is stored in: **data/faiss/index_info.json**

RAG Pipeline

The RAG pipeline consists of the following stages.

1. **Document Loading**
PDF documents are loaded from: data/documents/ using PyPDFDirectoryLoader.
2. **Document Splitting**
Documents are split using: RecursiveCharacterTextSplitter
With chunk_size = 1000, chunk_overlap = 200
The values are configurable through the application settings.
3. **Embeddings**
Each document chunk is converted into a vector using: BAAI/bge-small-en-v1.5

- The embedding model is loaded through HuggingFaceEmbeddings.
4. **FAISS Index**
The document vectors are stored in a local FAISS index.
The application loads an existing index when possible, rather than rebuilding it for every request.
 5. **Change Detection**
Before loading the existing index, the application checks:
 - Whether source documents changed
 - Whether the embedding model changed
 - Whether chunk size changed
 - Whether chunk overlap changedIf a change is detected, the FAISS index is rebuilt.
This makes the application more practical during development because the index does not have to be manually rebuilt whenever the knowledge base changes.
 6. **Retrieval**
The application creates a LangChain retriever with: k=4
Therefore, each query retrieves up to four document chunks.
 7. **Context Construction**
The retrieved document contents are concatenated into the context supplied to the LLM.
 8. **Prompt**
The system prompt instructs the model to:
 - Use only the provided context.
 - Accept semantic/synonym variations in the question.
 - Avoid unsupported inference.
 - Return a predefined response when the answer cannot be found.The fallback response is:
I couldn't find the answer in the provided documents.
 9. **LLM Generation**
The context and question are sent to: llama3.2:3b running locally through Ollama.
The application configures:
 - temperature
 - top-k
 - top-p
 - maximum generated tokensthrough environment-based settings.

Model Details

Embedding Model

Model: BAAI/bge-small-en-v1.5

Purpose: Convert enterprise document chunks and user questions into dense vector representations used for semantic retrieval.

The project does not fine-tune this embedding model in the current implementation.

LLM

Model: llama3.2:3b

Runtime: Ollama

Purpose: Generate the final answer from the retrieved enterprise context.

No LLM fine-tuning is performed in this version.

The model runs locally rather than through a paid hosted LLM API.

API

API	Purpose
-----	---------

/Get	Root. Returns a basic application message.
GET /health	Returns application health and Ollama availability. { "status": "healthy", "ollama": true}
GET /version	Returns: • application name • application version • LLM model • embedding model
POST /chat	Request: <pre>{ "question": "What are the company's revenue highlights?" }</pre> Response contains: • answer • sources • metadata Example metadata fields: <pre>{ "response_time_ms": 12345, "retrieved_chunks": 4, "source_documents": 2, "llm_model": "llama3.2:3b", "embedding_model": "BAAI/bge-small-en-v1.5", "vector_store": "FAISS" }</pre>
POST /reindex	Forces the FAISS vector index to be rebuilt.
GET /test-llm	Checks Ollama connectivity.
GET /metrics	Exposes Prometheus-compatible metrics.

Streamlit Application

The Streamlit application is intentionally separate from the Dockerized backend.

Start it from the project root: **uv run streamlit run streamlit_app.py**

Default URL: **http://localhost:8501**

The Streamlit application calls: **http://localhost:8000/chat**

The UI displays:

- Answer
- Sources
- Response time
- Retrieved chunks
- Number of source documents
- LLM model
- Embedding model
- Vector store
- Raw JSON response

The sidebar also provides links to:

- Grafana
- Prometheus
- Opik

Docker Deployment

- The FastAPI backend is containerized using the supplied Dockerfile.

- The Docker Compose deployment contains:
 - `enterprise-ai`
 - `prometheus`
 - `grafana`
- Ollama is deliberately not containerized.
- **Build and start**

```
docker compose up -d --build
```
- **Check services**
 - `docker compose ps`
- **View application logs**

```
docker logs enterprise-ai-container
```

or:

```
docker compose logs -f enterprise-ai
```
- **Stop services**

```
docker compose down
```
- Prometheus and Grafana use named Docker volumes so their data can survive container recreation.

Ollama Setup

- Ollama runs on the host machine.
- The Docker container connects to the host using:


```
host.docker.internal
```
- The application configuration uses an `OLLAMA_HOST` value pointing to the host Ollama service.
- Start Ollama on the host before using the RAG API.
- Verify the model is available:


```
ollama list
```
- The configured model is:


```
llama3.2:3b
```
- The FastAPI endpoint can also be used:


```
GET /test-llm
```

Prometheus

- Prometheus scrapes the FastAPI application at:


```
enterprise-ai:8000/metrics
```
- The current scrape interval is:


```
15 seconds
```
- Prometheus is exposed locally at:


```
http://localhost:9090
```
- **Application metrics**

The application exposes:

 - `rag_requests_total`
 - `rag_request_failures_total`
 - `rag_request_latency_seconds`
 - `llm_requests_total`
 - `llm_errors_total`
 - `retrieval_latency_seconds`
 - `llm_latency_seconds`
 - `retrieved_documents`
- In addition, the Prometheus Python client exposes standard Python/process metrics such as:
 - `python_gc_*`

- `python_info`
- `process_*`

Grafana

- Grafana is available at:
`http://localhost:3000`
- Configure Prometheus as the Grafana data source using:
`http://prometheus:9090`
- when Grafana is running in the same Docker Compose network.
- Recommended dashboard areas include:
 - Request volume
 - Request failures
 - Average RAG latency
 - P95 RAG latency
 - Average retrieval latency
 - P95 retrieval latency
 - Average LLM latency
 - P95 LLM latency
 - Retrieved chunks per request
- Example P95 LLM latency query:


```

      histogram_quantile(
        0.95,
        sum(rate(llm_latency_seconds_bucket[5m])) by (le)
      )
      
```
- Example average LLM latency:


```

      rate(llm_latency_seconds_sum[5m])
      /
      rate(llm_latency_seconds_count[5m])
      
```
- Example average retrieval latency:


```

      rate(retrieval_latency_seconds_sum[5m])
      /
      rate(retrieval_latency_seconds_count[5m])
      
```
- Example average RAG latency:


```

      rate(rag_request_latency_seconds_sum[5m])
      /
      rate(rag_request_latency_seconds_count[5m])
      
```
- For the retrieved_documents Summary metric, average retrieved chunks per request can be calculated as:


```

      rate(retrieved_documents_sum[5m])
      /
      rate(retrieved_documents_count[5m])
      
```

Opik

- Opik provides trace-level observability for individual RAG requests.
- Each request creates a trace:


```

      RAG Request
      
```
- with two major spans:


```

      RAG Request
      |--- Retrieval
      |___ Generation
      
```
- **Retrieval trace information**
- The Retrieval span records:

- question
- number of retrieved chunks
- source names
- document previews
- context size
- **Generation trace information**
 - The Generation span records:
 - question
 - system prompt
 - user prompt
 - context size
 - context chunk count
 - generated answer
 - LLM model
 - provider
- **Request-level metadata**
 - The RAG trace records:
 - embedding model
 - LLM model
 - vector store
 - top-k
 - chunk size
 - chunk overlap
 - documents retrieved
 - source documents
 - retrieval latency
 - LLM latency
 - total latency
- This makes Opik complementary to Prometheus and Grafana.
- Prometheus/Grafana answer:
 - How is the application performing over time?
- Opik answers:
 - What happened inside this particular RAG request?

Logging

- Application logging is implemented with Loguru.
- Console logs are written to standard output.
- File logs are stored under:
 - logs/
- The application configures:
 - logs/application.log
 - logs/errors.log
with rotation and retention policies.
- Third-party library logging is reduced to keep application logs readable.

Configuration

- Configuration is loaded using Pydantic Settings.
- The application reads environment variables from .env.
- Important settings include:

- LLM_MODEL
- OLLAMA_HOST
- EMBEDDING_MODEL
- HF_TOKEN
- TEMPERATURE
- TOP_K
- TOP_P
- MAX_TOKENS
- CHUNK_SIZE
- CHUNK_OVERLAP
- OPIK_API_KEY
- OPIK_WORKSPACE
- OPIK_PROJECT

Security

- Never commit: `.env` to source control.
- API keys, tokens, passwords, and other secrets should be provided through environment variables.

Project Structure

```
Enterprise_AI_Knowledge_Assistant/  
├── data/  
│   ├── documents/  
│   │   ├── BI_Glossary.pdf  
│   │   ├── Business_FAQ.pdf  
│   │   ├── Business_Policies.pdf  
│   │   ├── Company_Overview.pdf  
│   │   ├── Northwind_Annual_Report_2025.pdf  
│   │   └── Northwind_IT_Policy.pdf  
│   └── faiss/  
│       ├── index/  
│       │   ├── index.faiss  
│       │   └── index.pkl  
│       └── index_info.json  
├── docker/  
├── prometheus/  
│   └── prometheus.yml  
├── logs/  
├── src/  
│   ├── api.py  
│   ├── config.py  
│   ├── graph.py  
│   ├── llm.py  
│   ├── logger.py  
│   ├── metrics.py  
│   ├── models.py  
│   ├── observability.py  
│   ├── prompts.py  
│   ├── rag.py  
│   └── vector_store.py  
├── streamlit_app.py  
├── main.py  
├── Dockerfile  
├── docker-compose.yml  
├── prometheus/prometheus.yml  
├── pyproject.toml  
├── uv.lock  
├── .dockerignore  
└── .gitignore
```

Local Development

Prerequisites

Install:

- Python 3.11+
- uv
- Docker Desktop
- Ollama
- Git

Install dependencies

uv sync

Configure environment

Create .env with the required application settings.

Start Ollama

Start Ollama and ensure the configured model is available.

Start backend

Using Docker:

docker compose up -d --build

Start Streamlit

uv run streamlit run streamlit_app.py

Access URLs

Component	URL
Streamlit	http://localhost:8501
FastAPI	http://localhost:8000
Swagger	http://localhost:8000/docs
ReDoc	http://localhost:8000/redoc
Prometheus	http://localhost:9090
Grafana	http://localhost:3000
Opik	https://www.comet.com/opik

Request Flow

A typical question follows this sequence:

1. User enters question
2. Streamlit sends POST /chat
3. FastAPI validates request
4. RAG trace is started in Opik
5. FAISS retriever searches document vectors
6. Top 4 chunks are returned
7. Retrieved chunks are converted to context
8. Prompt is constructed
9. Ollama generates answer
10. Metrics are recorded
11. Trace is completed in Opik
12. FastAPI returns answer + sources + metadata
13. Streamlit displays response

Error Handling

The application handles several failure scenarios.

- **Ollama unavailable**
The /chat endpoint returns HTTP 503 when Ollama is unavailable.
- **Retrieval or generation failure**
The RAG pipeline increments:
`rag_request_failures_total`
and records error information in the Opik trace.
- **API connection failure**
The Streamlit application displays a user-friendly message when it cannot connect to FastAPI.
- **Request timeout**
Streamlit uses a 180-second HTTP timeout for RAG requests.

Index Management

The application automatically rebuilds the FAISS index when:

- the index does not exist
- source documents change
- embedding model changes
- chunk size changes
- chunk overlap changes

A manual rebuild can also be triggered using:

`POST /reindex`

This allows the knowledge base to evolve without manually deleting the vector index.

Current Limitations

The current version has several deliberate POC-level limitations:

1. FAISS is a local vector store rather than a distributed vector database.
2. Ollama runs locally and is not containerized.
3. Streamlit runs locally and is not containerized.
4. Authentication is not implemented.
5. Authorization at document level is not implemented.
6. No automated cloud deployment is included.
7. No automated horizontal scaling is implemented.
8. No automated model retraining pipeline is implemented.
9. Retrieval uses a single semantic retriever rather than hybrid retrieval.
10. No dedicated reranking model is used.
11. The current LLM is not fine-tuned.
12. Evaluation is currently primarily functional/observability oriented rather than a comprehensive automated RAG benchmark.

Challenges and Lessons Learned

- **Environment configuration**
The application uses environment-based configuration for model and observability settings.
- **Docker networking**
Ollama is hosted outside Docker, so the container must use `host.docker.internal` to reach the host service.
- **Persistent monitoring**
Prometheus and Grafana require persistent volumes if historical monitoring state is expected to survive container recreation.
- **RAG observability**
Aggregate metrics alone are insufficient for debugging an individual answer. Opik complements Prometheus/Grafana by showing the retrieval and generation steps for a specific request.
- **Development ergonomics**
Keeping Streamlit outside Docker makes UI changes faster during development.

Future Enhancements

Potential improvements include:

1. Hybrid dense + BM25 retrieval.
2. Cross-encoder reranking.
3. Embedding model fine-tuning.
4. Automated RAG evaluation using a golden dataset.
5. Faithfulness and relevance evaluation.
6. Token-level LLM metrics.
7. Alerting in Grafana.
8. Kubernetes deployment.
9. Horizontal scaling.
10. Authentication and RBAC.
11. Managed vector database.
12. Production model serving.
13. CI/CD.
14. Automated model lifecycle management.

Conclusion

The Enterprise AI Knowledge Assistant demonstrates the complete lifecycle of a local LLM-powered RAG proof of concept. It combines retrieval, generation, API serving, UI, containerization, metrics, dashboards, traces, and logging into one solution.

The key LLMOps principle demonstrated by the project is that deploying an LLM application is not limited to generating an answer. The application must also provide visibility into:

- request volume
- failures
- retrieval performance
- generation performance
- retrieved content
- individual traces
- application health

Prometheus, Grafana, and Opik provide complementary visibility into these areas.

ANNEXURE – APPLICATION SCREENSHOTS

System

LLM

llama3.2:3b

Embedding Model

BAAI/bge-small-en-v1.5

Vector Store

FAISS

Backend

FastAPI

Tracing

Opik

Monitoring

Prometheus

Visualization

Grafana

Observability

Enterprise AI Knowledge Assistant

Ask questions about enterprise documents using a Retrieval-Augmented Generation (RAG) pipeline.

Question

What is Returns, Claims & Credit Refund Policy at Northwind?

Sample Questions

Choose a sample question

Choose an option

Ask

Clear

Answer

The Returns, Claims & Credit Refund Policy at Northwind Traders states that validated claims receive a Credit Memo applied to the customer's account balance.

Sources

data\documents\Business_FAQ.pdf

data\documents\Business_Policies.pdf

Vector Store

FAISS

Backend

FastAPI

Tracing

Opik

Monitoring

Prometheus

Visualization

Grafana

Observability

Metadata

Response Time: 53055 ms

Retrieved Chunks: 4

Source Documents: 2

LLM Model: llama3.2:3b

Embedding Model: BAAI/bge-small-en-v1.5

Vector Store: FAISS

Raw JSON Response

```
{  "answer": "The Returns, Claims & Credit Refund Policy at Northwind Traders states that validated claims receive a Credit Memo applied to the customer's account balance.",  "sources": [    {      "source": "data\\documents\\Business_FAQ.pdf"    },    {      "source": "data\\documents\\Business_Policies.pdf"    }  ],  "metadata": {    "response_time_ms": 53055,    "retrieved_chunks": 4,    "source_documents": 2,    "llm_model": "llama3.2:3b",    "embedding_model": "BAAI/bge-small-en-v1.5",    "vector_store": "FAISS"  }}
```

LLM

llama3.2:3b

Embedding Model

BAAI/bge-small-en-v1.5

Vector Store

FAISS

Backend

FastAPI

Tracing

Opik

Monitoring

Prometheus

Visualization

Grafana

Observability

Open Grafana

Ask questions about enterprise documents using a Retrieval-Augmented Generation (RAG) pipeline.

Question

Give me summary of Northwind's Annual Report 2025

Sample Questions

Choose a sample question

Choose an option

AskClear

Answer

Northwind Traders' Annual Report 2025 highlights a period of robust operational consolidation and growth. The company achieved:

- \$14.85M in total gross revenue, a 12.4% year-over-year increase
- Expanded net profit margins by 180 basis points, driven by logistics optimization and enhanced bulk procurement contracts
- Successfully integrated automated inventory telemetry across 8 central logistics hubs, reducing stockouts for high-demand items by 28%
- Strong performance in regional markets, with North America and Europe contributing 87% of total revenue
- Improved freight and carrier operational performance, including a Freight-to-Goods Value Ratio of 5.65%

The report also outlines the company's technology strategy, including the development of Next-Gen SQL Analytics and RAG knowledge retrieval pipeline.

Sources

LLM

llama3.2:3b

Embedding Model

BAAI/bge-small-en-v1.5

Vector Store

FAISS

Backend

FastAPI

Tracing

Opik

Monitoring

Prometheus

Visualization

Grafana

Observability

Open Grafana

Ask questions about enterprise documents using a Retrieval-Augmented Generation (RAG) pipeline.

Question

Example: What are the company's revenue highlights?

Sample Questions

Choose a sample question

Summarize the company overview.

AskClear

Answer

Northwind Traders is a company that achieved robust operational consolidation and growth in Fiscal Year 2025, driven by strong demand in core European and North American markets. The company expanded its gross revenue by 12.4% year-over-year to \$14.85 Million, with net profit margins increasing by 180 basis points due to logistics optimization and enhanced bulk procurement contracts.

Sources

data\documents\Business_FAQ.pdf

data\documents\Northwind_Annual_Report_2025.pdf

data\documents\BI_Glossary.pdf

FASTAPI METRICS - Metadata – Retrieval and Performance Details

← ↻ ⓘ localhost:8000/metrics

```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 57143.0
python_gc_objects_collected_total{generation="1"} 5481.0
python_gc_objects_collected_total{generation="2"} 477.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 1797.0
python_gc_collections_total{generation="1"} 163.0
python_gc_collections_total{generation="2"} 10.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="16",version="3.11.16"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 9.794756608e+09
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 1.172185088e+09
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.78677551841e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 65.4
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 19.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.048576e+06
# HELP rag_requests_total Total RAG requests
# TYPE rag_requests_total counter
rag_requests_total 6.0
# HELP rag_requests_created Total RAG requests
# TYPE rag_requests_created gauge
rag_requests_created 1.7867755247739089e+09
# HELP rag_request_failures_total Failed RAG requests
# TYPE rag_request_failures_total counter
rag_request_failures_total 0.0
# HELP rag_request_failures_created Failed RAG requests
# TYPE rag_request_failures_created gauge
rag_request_failures_created 1.7867755247739253e+09
# HELP rag_request_latency_seconds RAG request latency
# TYPE rag_request_latency_seconds histogram
rag_request_latency_seconds_bucket{le="0.005"} 0.0
rag_request_latency_seconds_bucket{le="0.01"} 0.0
rag_request_latency_seconds_bucket{le="0.025"} 0.0
rag_request_latency_seconds_bucket{le="0.05"} 0.0
rag_request_latency_seconds_bucket{le="0.075"} 0.0
rag_request_latency_seconds_bucket{le="0.1"} 0.0
rag_request_latency_seconds_bucket{le="0.25"} 0.0
rag_request_latency_seconds_bucket{le="0.5"} 0.0
rag_request_latency_seconds_bucket{le="0.75"} 0.0
rag_request_latency_seconds_bucket{le="1.0"} 0.0
rag_request_latency_seconds_bucket{le="2.5"} 0.0
rag_request_latency_seconds_bucket{le="5.0"} 1.0
rag_request_latency_seconds_bucket{le="7.5"} 4.0
rag_request_latency_seconds_bucket{le="10.0"} 4.0
```


PROMETHEUS – Metrics

 Prometheus

Query

Alerts

Status ▾



>_ rag_requests_total

:

Execute

Table Graph Explain

< Evaluation time >

Load time: 17ms Result series: 1

rag_requests_total(instance="enterprise-ai:8000",job="enterprise-ai")

2

 Prometheus

Query

Alerts

Status ▾



>_ rag_request_failures_total

:

Execute

Table Graph Explain

< Evaluation time >

Load time: 16ms Result series: 1

rag_request_failures_total(instance="enterprise-ai:8000",job="enterprise-ai")

0

>_
 rate(rag_request_latency_seconds_sum[5m])
 /
 rate(rag_request_latency_seconds_count[5m])

:

Execute

Table Graph Explain

< Evaluation time >

Load time: 14ms Result series: 1

(instance="enterprise-ai:8000",job="enterprise-ai")

5.730425083

>_
 rate(retrieval_latency_seconds_sum[5m])
 /
 rate(retrieval_latency_seconds_count[5m])

:

Execute

Table Graph Explain

< Evaluation time >

Load time: 13ms Result series: 1

(instance="enterprise-ai:8000",job="enterprise-ai")

0.08157133400004568

GRAFANA – Monitoring Dashboard

localhost:3000/d/ad99jvb/enterprise-ai-knowledge-assistant?orgId=1&from=now-6h&to=now&timezone=browser

Grafana Dashboards > Enterprise AI Knowledge Assistant

- Home
- Bookmarks
- Starred
- Dashboards
- Playlists
- Snapshots
- Library panels
- Shared dashboards
- Recently deleted
- Explore
- Drilldown
- Alerting
- Connections
- Add new connection
- Data sources

Overview

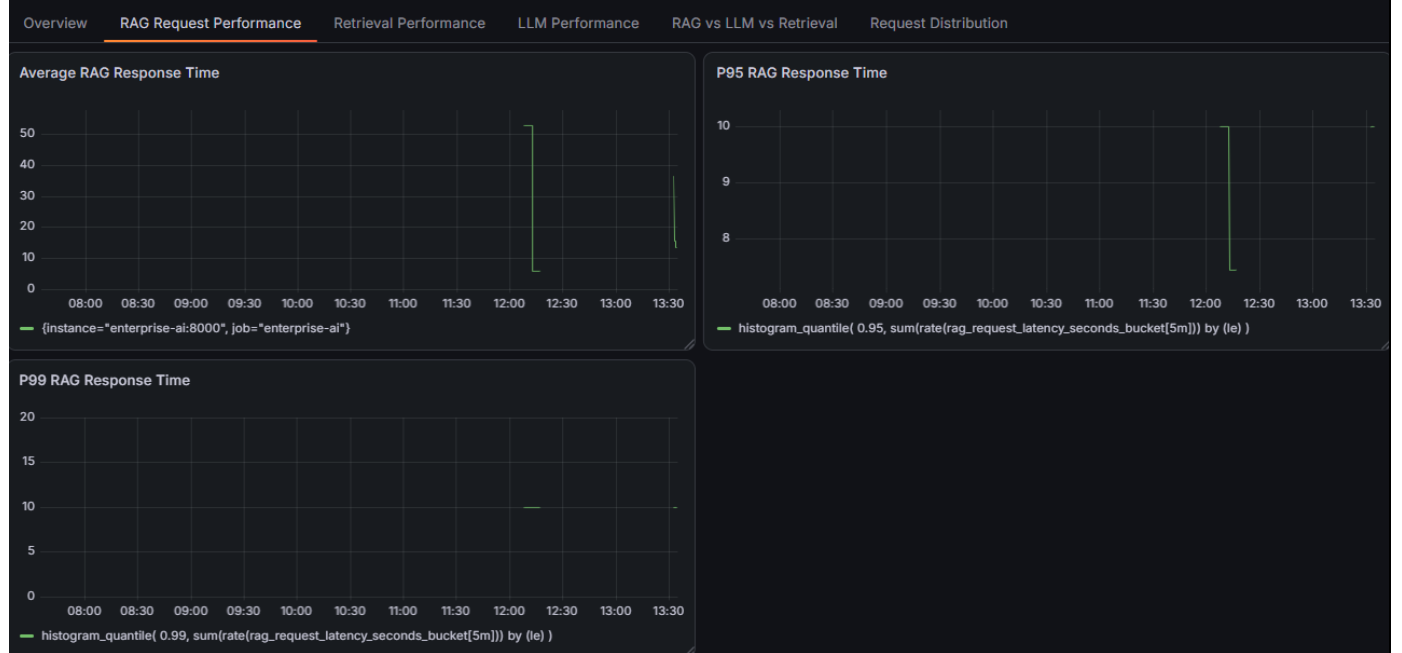
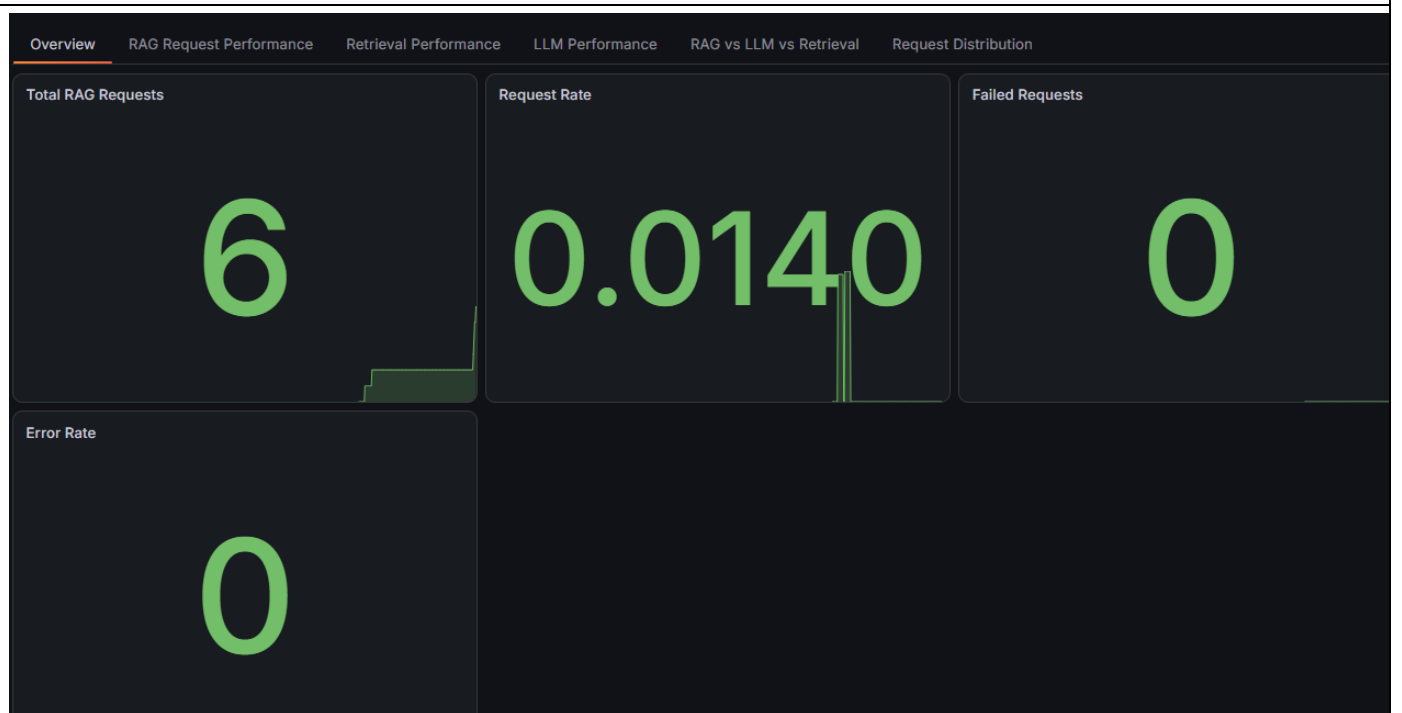
RAG Request Performance

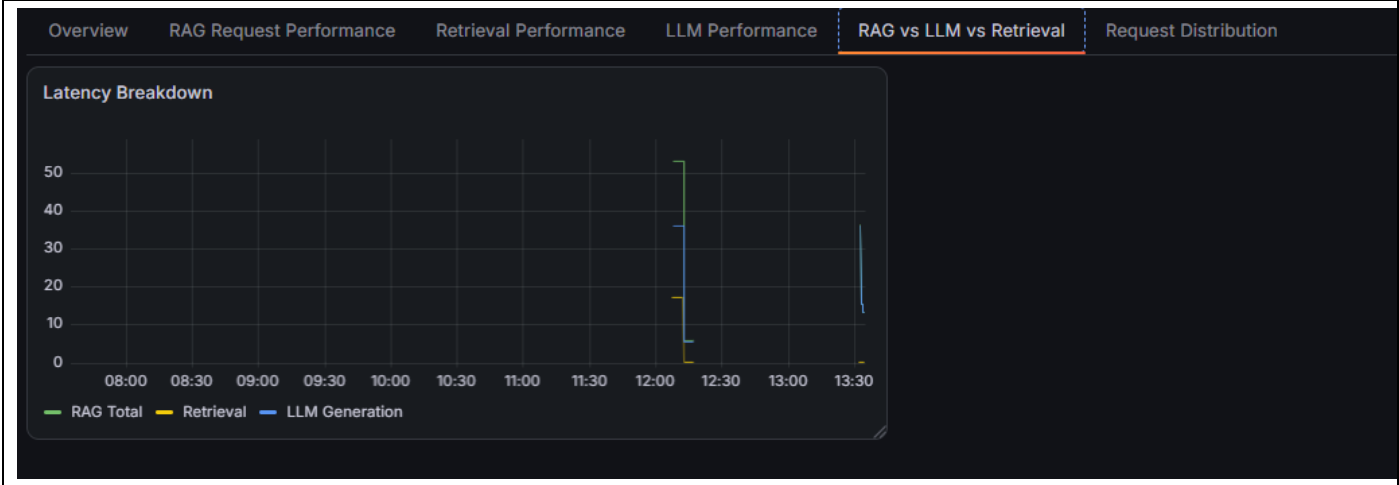
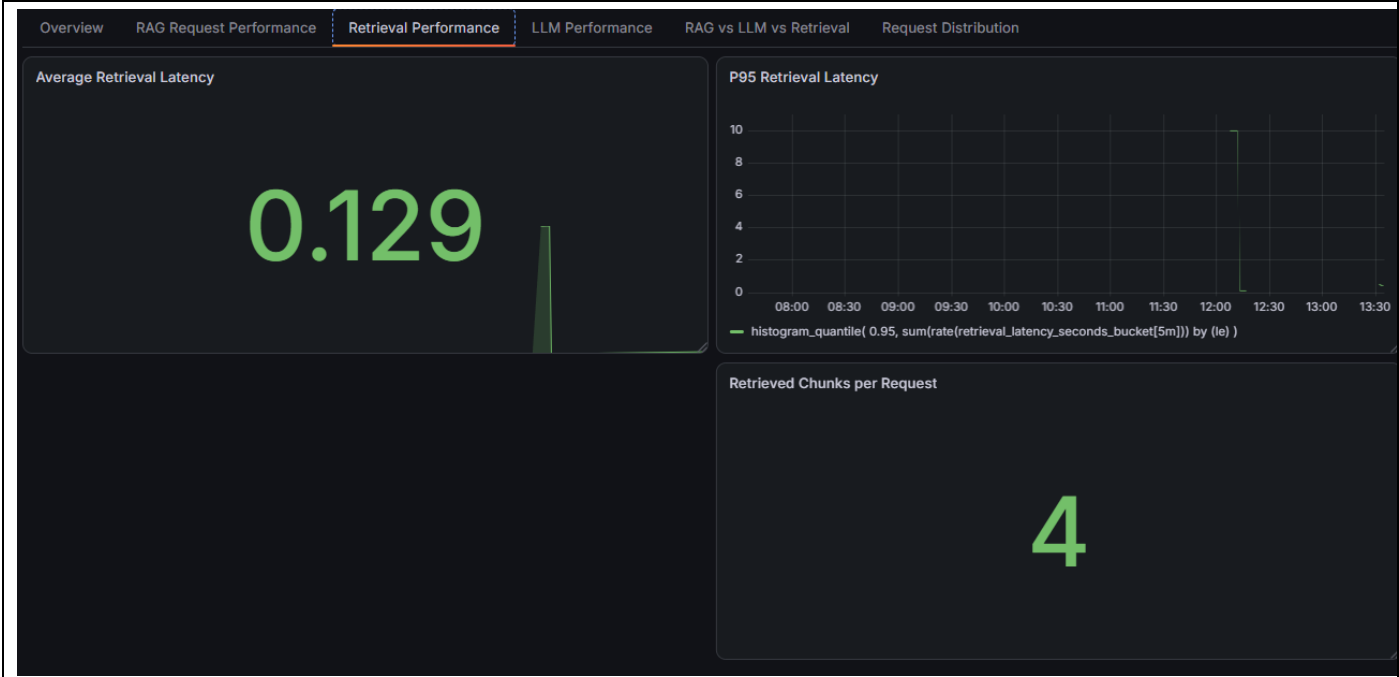
Retrieval Performance

LLM Performance

RAG vs LLM vs Retrieval

Request Distribution





Overview

RAG Request Performance

Retrieval Performance

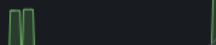
LLM Performance

RAG vs LLM vs Retrieval

Request Distribution

Requests per Minute

0.842



OPIK – LLM Trace

opik

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akanksha-patil-0279 / Enterprise AI Knowledge Assistant / Logs

Upgrade

Logs

Set a guardrail

ThreadsTracesSpansColumns 10/22All environmentsPast 30 days

{ } Traces 15

Error rate 0% No changes

Avg duration 16.9s

Total cost \$0 No changes

8

0

07/1707/1807/1907/2007/2107/2207/2307/2407/2507/2607/2707/2807/2907/3007/3108/0108/0308/0508/0708/0908/1008/1108/1208/1308/1408/15

Search by anything

With errors

Tags

Metadata

All filters

	Start time	Input count 14	Output count 15	Errors count 0
☐	15 Aug 2026, 1:33 PM	Give me summary of Northwind's Annual Report 2025	Northwind Traders' Annual Report 2025 highlights a period of robust opera...	
☐	15 Aug 2026, 1:32 PM	What products does Northwind sell?	Northwind sells beverages, condiments, confections, dairy products, grain...	
☐	15 Aug 2026, 1:32 PM	Explain the business glossary.	The Business Intelligence Glossary serves as a single source of truth for...	
☐	15 Aug 2026, 1:31 PM	What are the key financial highlights?	The key financial highlights include: * Total gross revenue of \$14.85 Mil...	
☐	15 Aug 2026, 12:12 ...	Summarize the company overview.	Northwind Traders is a company that achieved robust operational consolida...	
☐	15 Aug 2026, 12:06 ...	What is Returns, Claims & Credit Refund Policy at NL	The Returns, Claims & Credit Refund Policy at Northwind Traders states th...	
☐	14 Aug 2026, 3:16 PM	What products does Northwind sell?	Northwind sells beverages, condiments, confections, dairy products, grain...	
☐	14 Aug 2026, 2:06 P...	Explain the business glossary.	The Business Intelligence Glossary serves as a single source of truth for...	

» RAG Request

PreviousNext

Spans (2)

RAG Request

4.1s

Retrieval

0.03s

Generation

4s

ollama llama3.2:3b

Inspect: RAG Request

Add toAnnotate

15 Aug 2026, 1:32 PM

4.1s

+

Details

Feedback scores

Input

Pretty

What products does Northwind sell?

Output

Pretty

Northwind sells beverages, condiments, confections, dairy products, grains/cereals, meat/poultry, produce, and seafood.

Metadata

YAML

```
providers:
  - ollama
embedding_model: BAAI/bge-small-en-v1.5
llm_model: llama3.2:3b
vector_store: FAISS
top_k: 4
chunk_size: 1000
chunk_overlap: 200
documents_retrieved: 4
source_documents: 1
sources:
  - data\documents\Business_FAQ.pdf
retrieval_latency_seconds: 0.031
llm_latency_seconds: 4.016
total_latency_seconds: 4.051
```

opik

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akanksha-patil-0279 / Enterprise AI Knowledge Assistant / Logs

Upgrade

Logs

ThreadsTracesSpans

Columns 11/19

All environments

Past 30 days

Set a guardrail

{ } Spans 28

Error rate 0% No changes

Avg duration 8.5s

Total cost \$0 No changes

16

0

07/1707/1807/1907/2007/2107/2207/2307/2407/2507/2607/2707/2807/2907/3007/3108/0108/0308/0508/0708/0808/0908/1008/1108/1208/1308/1408/15

Search by anything

Type

Tags

With errors

Metadata

All filters

☐	Start time	Input	Output	Errors	Name
	-	count 28	count 28	count 0	-
☐	15 Aug 2026, 1:33 PM	Give me summary of Northwnd's Annual Report 2025	Northwind Traders' Annual Report 2025 highlights a ...		Generatio
☐	15 Aug 2026, 1:33 PM	Give me summary of Northwnd's Annual Report 2025	{ "documents_retrieved": 4, "sources": ["data\doc...		Retrieval
☐	15 Aug 2026, 1:32 PM	What products does Northwnd sell?	Northwind sells beverages, condiments, confections,...		Generatio
☐	15 Aug 2026, 1:32 PM	What products does Northwnd sell?	{ "documents_retrieved": 4, "sources": ["data\doc...		Retrieval
☐	15 Aug 2026, 1:32 PM	Explain the business glossary.	The Business Intelligence Glossary serves as a sing...		Generatio
☐	15 Aug 2026, 1:32 PM	Explain the business glossary.	{ "documents_retrieved": 4, "sources": ["data\doc...		Retrieval
☐	15 Aug 2026, 1:31 PM	What are the key financial highlights?	The key financial highlights include: * Total gross...		Generatio
☐	15 Aug 2026, 1:31 PM	What are the key financial highlights?	{ "documents_retrieved": 4, "sources": ["data\doc...		Retrieval

DOCKER - Running Containers / images

☐	Name	Tag	Image ID	Creat... ↓	Size	Actions
☐	enterprise-ai	latest	1a492053c63b	2 hours ago	5.8 GB	
☐	grafana/grafana	latest	8b5e93ab63f5	8 days ago	1.15 GB	
☐	prom/prometheus	latest	8da6d95a8747	16 days ago	250.29 MB	

Containers [Give feedback](#)

Container CPU usage ⓘ
1.84% / 1200% (12 CPUs available)

Container memory usage ⓘ
1.04GB / 7.53GB

Show charts

☐ Only show running containers

Delete

☒	Name	Container ID	Image	Port(s)	CPU (%)	Actions
☒	4_enterprise_ai_knowledge_assistant	-	-	-	1.84%	